AR-601M: analytical inverse kinematics

Task 3 derivation for the right and left arms

1. Model and target

Let $b = +1$ for the right arm and $b = -1$ for the left arm. R_a and D_a denote homogeneous rotation and translation about or along the *current local* axis a. The forward chain is

$$\begin{aligned} T_{0t} = {} & R_z(q_1)D_z(l_1)R_z(b\alpha)R_y(q_2)D_y(-bl_2) \\ & R_x(q_3)D_z(-l_3)R_z(q_4)R_y(q_5)R_x(q_6)D_x(l_4), \end{aligned} \tag{1}$$

where $\alpha = 10^\circ$ and $(l_1, l_2, l_3, l_4) = (0.40, 0.18, 0.30, 0.12)$ m. Angles are evaluated in radians. The joint axes are z, y, x, z, y, x in that order. Let the requested pose be $T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \in SE(3)$, with e_x, e_y, e_z the standard unit vectors. Remove the final tool offset first:

$$W = p - l_4 R e_x, \qquad C = l_1 e_z, \qquad r = W - C = (x, y, z)^\mathsf{T}, \qquad \theta = q_1 + b\alpha. \tag{2}$$

W is the common wrist-joint centre; C is the end of the first offset. Because $R_y(q_2)e_y = e_y$, the end of the shoulder offset is

$$S = C - bl_2 R_z(\theta) e_y. \tag{3}$$

2. Solve the shoulder rotation q_1

The segment from S to W has length l_3. Expanding its squared length, using $R_z(\theta)e_y = (-\sin\theta, \cos\theta, 0)^\mathsf{T}$, gives

$$\begin{aligned} l_3^2 &= \|W - S\|^2 = \|r + bl_2 R_z(\theta) e_y\|^2 \\ &= \|r\|^2 + l_2^2 - 2bl_2(x\sin\theta - y\cos\theta). \end{aligned} \tag{4}$$

Set $K = (\|r\|^2 + l_2^2 - l_3^2)/(2l_2)$, $\rho = \sqrt{x^2 + y^2}$, and $\phi = \operatorname{atan2}(y, x)$. Since $b^{-1} = b$,

$$x\sin\theta - y\cos\theta = \rho\sin(\theta - \phi) = bK. \tag{5}$$

For $\rho > 0$ and $|K| \le \rho$, the two candidates are

$$\theta = \phi + \arcsin(bK/\rho) \quad \text{or} \quad \theta = \phi + \pi - \arcsin(bK/\rho), \qquad q_1 = \theta - b\alpha \pmod{2\pi}. \tag{6}$$

They coincide modulo 2π when $|K| = \rho$. If $|K| > \rho$, the target is unreachable. If $\rho = 0$, equation (5) requires $K = 0$; then q_1 is free and the remaining joints must be solved for each chosen value. These rules apply to both arms: changing b changes the signs of the 10° rotation and the shoulder offset.

3. Solve the arm joints q_2, q_3

For each q_1 candidate, express the wrist displacement in the axes after the fixed rotation. Equations (1)–(3) give

$$u = R_z(-\theta)r + bl_2 e_y = R_y(q_2)R_x(q_3)(-l_3 e_z) = l_3 \begin{bmatrix} -\sin q_2 \cos q_3 \\ \sin q_3 \\ -\cos q_2 \cos q_3 \end{bmatrix}. \tag{7}$$

For each $\sigma \in \{+1, -1\}$, representing the sign of $\cos q_3$,

$$c = \sqrt{l_3^2 - u_y^2}, \qquad q_3 = \operatorname{atan2}(u_y, \sigma c), \qquad q_2 = \operatorname{atan2}(-\sigma u_x, -\sigma u_z). \tag{8}$$

When $c = 0$, $q_3 = \pm 90^\circ$ and q_2 is free. This is a continuous family before limits, but it violates the specified $q_3 \in [-45^\circ, 45^\circ]$ limit.

4. Solve the spherical wrist q_4, q_5, q_6

The arm orientation before the wrist is $A = R_z(\theta)R_y(q_2)R_x(q_3)$. Hence

$$Q = A^{\mathsf{T}}R = R_z(q_4)R_y(q_5)R_x(q_6). \tag{9}$$

Its entries satisfy $Q_{11} = \cos q_4 \cos q_5$, $Q_{21} = \sin q_4 \cos q_5$, $Q_{31} = -\sin q_5$, $Q_{32} = \cos q_5 \sin q_6$, and $Q_{33} = \cos q_5 \cos q_6$. Put $h = \sqrt{Q_{11}^2 + Q_{21}^2} = |\cos q_5|$. For $h > 0$, each $\tau \in \{+1, -1\}$ gives one Euler branch:

$$\begin{aligned} q_5 &= \operatorname{atan2}(-Q_{31}, \tau h), \\ q_4 &= \operatorname{atan2}(\tau Q_{21}, \tau Q_{11}), \\ q_6 &= \operatorname{atan2}(\tau Q_{32}, \tau Q_{33}). \end{aligned} \tag{10}$$

If $h = 0$, let $s = -Q_{31} \in \{+1, -1\}$ and $\delta = \operatorname{atan2}(sQ_{23}, Q_{22})$. Then

$$q_5 = s\pi/2, \qquad q_4 - sq_6 = \delta \pmod{2\pi}. \tag{11}$$

This is a continuous wrist family, not an isolated branch. Its feasible q_6 values satisfy $q_6 \in [L_6, U_6]$ and, for some integer k, $L_4 \le \delta + sq_6 + 2\pi k \le U_4$.

5. Branch selection and checks

For a nonsingular target, equations (6), (8), and (10) produce at most $2 \cdot 2 \cdot 2 = 8$ real branches, for either value of b. Angles are wrapped to $[-\pi, \pi)$. Two branches are duplicates when $\max_i |\operatorname{wrap}(q_i - \tilde{q}_i)| < 10^{-8}$ rad; duplicates are removed *before* applying the closed joint limits. Limit comparisons allow 10^{-9} rad for rounding at a bound. Each isolated branch must reproduce T with $\|FK(q) - T\|_F \le 10^{-8}$. The implementation also checks that the input is a valid pose in $SE(3)$ and uses 10^{-10} m for positional degeneracy tests.

The implementation is `src/task3_ik.py`; numerical branch lists, feasibility, and position/orientation residuals for both arms are in `REPORT.md`, Task 3.